

FRAME DEMONSTRATION

SSS 1-3 | Advanced Production Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for production and resource-use efficiency.

DEMONSTRATE

Demonstrate how to frame using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-401 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

DESIGN DEMONSTRATION

SSS 1-3 | Advanced Production Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for production and resource-use efficiency.

DEMONSTRATE

Demonstrate how to design using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-402 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

INSTRUMENT DEMONSTRATION

SSS 1-3 | Advanced Production Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for production and resource-use efficiency.

DEMONSTRATE

Demonstrate how to instrument using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

CALIBRATE DEMONSTRATION

SSS 1-3 | Advanced Production Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for production and resource-use efficiency.

DEMONSTRATE

Demonstrate how to calibrate using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-404 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

MODEL DEMONSTRATION

SSS 1-3 | Advanced Production Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for production and resource-use efficiency.

DEMONSTRATE

Demonstrate how to model using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-405 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

VALIDATE DEMONSTRATION

SSS 1-3 | Advanced Production Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for production and resource-use efficiency.

DEMONSTRATE

Demonstrate how to validate using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-406 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

EVALUATE DEMONSTRATION

SSS 1-3 | Advanced Production Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for production and resource-use efficiency.

DEMONSTRATE

Demonstrate how to evaluate using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-407 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

OPTIMIZE DEMONSTRATION

SSS 1-3 | Advanced Production Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for production and resource-use efficiency.

DEMONSTRATE

Demonstrate how to optimize using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-408 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

STRESS-TEST DEMONSTRATION

SSS 1-3 | Advanced Production Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for production and resource-use efficiency.

DEMONSTRATE

Demonstrate how to stress-test using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-409 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

RECOMMEND DEMONSTRATION

SSS 1-3 | Advanced Production Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for production and resource-use efficiency.

DEMONSTRATE

Demonstrate how to recommend using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-410 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

FRAME DEMONSTRATION

SSS 1-3 | Crop Trial Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for replicated treatment trials.

DEMONSTRATE

Demonstrate how to frame using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-411 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

DESIGN DEMONSTRATION

SSS 1-3 | Crop Trial Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for replicated treatment trials.

DEMONSTRATE

Demonstrate how to design using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-412 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

INSTRUMENT DEMONSTRATION

SSS 1-3 | Crop Trial Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for replicated treatment trials.

DEMONSTRATE

Demonstrate how to instrument using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-413 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

CALIBRATE DEMONSTRATION

SSS 1-3 | Crop Trial Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for replicated treatment trials.

DEMONSTRATE

Demonstrate how to calibrate using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-414 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

MODEL DEMONSTRATION

SSS 1-3 | Crop Trial Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for replicated treatment trials.

DEMONSTRATE

Demonstrate how to model using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-415 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

VALIDATE DEMONSTRATION

SSS 1-3 | Crop Trial Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for replicated treatment trials.

DEMONSTRATE

Demonstrate how to validate using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-416 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

EVALUATE DEMONSTRATION

SSS 1-3 | Crop Trial Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for replicated treatment trials.

DEMONSTRATE

Demonstrate how to evaluate using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-417 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

OPTIMIZE DEMONSTRATION

SSS 1-3 | Crop Trial Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for replicated treatment trials.

DEMONSTRATE

Demonstrate how to optimize using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-418 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

STRESS-TEST DEMONSTRATION

SSS 1-3 | Crop Trial Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for replicated treatment trials.

DEMONSTRATE

Demonstrate how to stress-test using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

RECOMMEND DEMONSTRATION

SSS 1-3 | Crop Trial Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for replicated treatment trials.

DEMONSTRATE

Demonstrate how to recommend using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-420 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

FRAME DEMONSTRATION

SSS 1-3 | Controlled Experiment Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for causal experimental design.

DEMONSTRATE

Demonstrate how to frame using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-421 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

DESIGN DEMONSTRATION

SSS 1-3 | Controlled Experiment Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for causal experimental design.

DEMONSTRATE

Demonstrate how to design using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-422 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

INSTRUMENT DEMONSTRATION

SSS 1-3 | Controlled Experiment Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for causal experimental design.

DEMONSTRATE

Demonstrate how to instrument using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-423 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

CALIBRATE DEMONSTRATION

SSS 1-3 | Controlled Experiment Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for causal experimental design.

DEMONSTRATE

Demonstrate how to calibrate using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

MODEL DEMONSTRATION

SSS 1-3 | Controlled Experiment Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for causal experimental design.

DEMONSTRATE

Demonstrate how to model using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-425 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

VALIDATE DEMONSTRATION

SSS 1-3 | Controlled Experiment Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for causal experimental design.

DEMONSTRATE

Demonstrate how to validate using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-426 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

EVALUATE DEMONSTRATION

SSS 1-3 | Controlled Experiment Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for causal experimental design.

DEMONSTRATE

Demonstrate how to evaluate using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-427 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

OPTIMIZE DEMONSTRATION

SSS 1-3 | Controlled Experiment Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for causal experimental design.

DEMONSTRATE

Demonstrate how to optimize using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

STRESS-TEST DEMONSTRATION

SSS 1-3 | Controlled Experiment Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for causal experimental design.

DEMONSTRATE

Demonstrate how to stress-test using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

RECOMMEND DEMONSTRATION

SSS 1-3 | Controlled Experiment Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for causal experimental design.

DEMONSTRATE

Demonstrate how to recommend using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-430 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

FRAME DEMONSTRATION

SSS 1-3 | Irrigation Engineering Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for hydraulic performance.

DEMONSTRATE

Demonstrate how to frame using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-431 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

DESIGN DEMONSTRATION

SSS 1-3 | Irrigation Engineering Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for hydraulic performance.

DEMONSTRATE

Demonstrate how to design using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-432 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

INSTRUMENT DEMONSTRATION

SSS 1-3 | Irrigation Engineering Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for hydraulic performance.

DEMONSTRATE

Demonstrate how to instrument using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

CALIBRATE DEMONSTRATION

SSS 1-3 | Irrigation Engineering Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for hydraulic performance.

DEMONSTRATE

Demonstrate how to calibrate using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

MODEL DEMONSTRATION

SSS 1-3 | Irrigation Engineering Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for hydraulic performance.

DEMONSTRATE

Demonstrate how to model using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-435 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

VALIDATE DEMONSTRATION

SSS 1-3 | Irrigation Engineering Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for hydraulic performance.

DEMONSTRATE

Demonstrate how to validate using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-436 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

EVALUATE DEMONSTRATION

SSS 1-3 | Irrigation Engineering Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for hydraulic performance.

DEMONSTRATE

Demonstrate how to evaluate using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-437 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

OPTIMIZE DEMONSTRATION

SSS 1-3 | Irrigation Engineering Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for hydraulic performance.

DEMONSTRATE

Demonstrate how to optimize using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-438 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

STRESS-TEST DEMONSTRATION

SSS 1-3 | Irrigation Engineering Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for hydraulic performance.

DEMONSTRATE

Demonstrate how to stress-test using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

RECOMMEND DEMONSTRATION

SSS 1-3 | Irrigation Engineering Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for hydraulic performance.

DEMONSTRATE

Demonstrate how to recommend using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-440 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

FRAME DEMONSTRATION

SSS 1-3 | Water Productivity Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for water balance and productivity.

DEMONSTRATE

Demonstrate how to frame using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-441 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

DESIGN DEMONSTRATION

SSS 1-3 | Water Productivity Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for water balance and productivity.

DEMONSTRATE

Demonstrate how to design using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-442 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

INSTRUMENT DEMONSTRATION

SSS 1-3 | Water Productivity Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for water balance and productivity.

DEMONSTRATE

Demonstrate how to instrument using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

CALIBRATE DEMONSTRATION

SSS 1-3 | Water Productivity Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for water balance and productivity.

DEMONSTRATE

Demonstrate how to calibrate using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-444 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

MODEL DEMONSTRATION

SSS 1-3 | Water Productivity Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for water balance and productivity.

DEMONSTRATE

Demonstrate how to model using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

VALIDATE DEMONSTRATION

SSS 1-3 | Water Productivity Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for water balance and productivity.

DEMONSTRATE

Demonstrate how to validate using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-446 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

EVALUATE DEMONSTRATION

SSS 1-3 | Water Productivity Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for water balance and productivity.

DEMONSTRATE

Demonstrate how to evaluate using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-447 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

OPTIMIZE DEMONSTRATION

SSS 1-3 | Water Productivity Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for water balance and productivity.

DEMONSTRATE

Demonstrate how to optimize using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

STRESS-TEST DEMONSTRATION

SSS 1-3 | Water Productivity Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for water balance and productivity.

DEMONSTRATE

Demonstrate how to stress-test using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

RECOMMEND DEMONSTRATION

SSS 1-3 | Water Productivity Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for water balance and productivity.

DEMONSTRATE

Demonstrate how to recommend using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-450 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

FRAME DEMONSTRATION

SSS 1-3 | Advanced Soilless Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for recirculating soilless control.

DEMONSTRATE

Demonstrate how to frame using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-451 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

DESIGN DEMONSTRATION

SSS 1-3 | Advanced Soilless Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for recirculating soilless control.

DEMONSTRATE

Demonstrate how to design using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-452 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

INSTRUMENT DEMONSTRATION

SSS 1-3 | Advanced Soilless Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for recirculating soilless control.

DEMONSTRATE

Demonstrate how to instrument using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-453 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

CALIBRATE DEMONSTRATION

SSS 1-3 | Advanced Soilless Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for recirculating soilless control.

DEMONSTRATE

Demonstrate how to calibrate using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-454 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

MODEL DEMONSTRATION

SSS 1-3 | Advanced Soilless Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for recirculating soilless control.

DEMONSTRATE

Demonstrate how to model using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-455 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

VALIDATE DEMONSTRATION

SSS 1-3 | Advanced Soilless Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for recirculating soilless control.

DEMONSTRATE

Demonstrate how to validate using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-456 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

EVALUATE DEMONSTRATION

SSS 1-3 | Advanced Soilless Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for recirculating soilless control.

DEMONSTRATE

Demonstrate how to evaluate using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

OPTIMIZE DEMONSTRATION

SSS 1-3 | Advanced Soilless Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for recirculating soilless control.

DEMONSTRATE

Demonstrate how to optimize using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

STRESS-TEST DEMONSTRATION

SSS 1-3 | Advanced Soilless Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for recirculating soilless control.

DEMONSTRATE

Demonstrate how to stress-test using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

RECOMMEND DEMONSTRATION

SSS 1-3 | Advanced Soilless Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for recirculating soilless control.

DEMONSTRATE

Demonstrate how to recommend using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-460 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

FRAME DEMONSTRATION

SSS 1-3 | Sensor & Monitoring Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for calibration and monitoring quality.

DEMONSTRATE

Demonstrate how to frame using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-461 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

DESIGN DEMONSTRATION

SSS 1-3 | Sensor & Monitoring Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for calibration and monitoring quality.

DEMONSTRATE

Demonstrate how to design using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-462 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

INSTRUMENT DEMONSTRATION

SSS 1-3 | Sensor & Monitoring Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for calibration and monitoring quality.

DEMONSTRATE

Demonstrate how to instrument using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-463 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

CALIBRATE DEMONSTRATION

SSS 1-3 | Sensor & Monitoring Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for calibration and monitoring quality.

DEMONSTRATE

Demonstrate how to calibrate using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

MODEL DEMONSTRATION

SSS 1-3 | Sensor & Monitoring Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for calibration and monitoring quality.

DEMONSTRATE

Demonstrate how to model using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-465 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

VALIDATE DEMONSTRATION

SSS 1-3 | Sensor & Monitoring Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for calibration and monitoring quality.

DEMONSTRATE

Demonstrate how to validate using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-466 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

EVALUATE DEMONSTRATION

SSS 1-3 | Sensor & Monitoring Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for calibration and monitoring quality.

DEMONSTRATE

Demonstrate how to evaluate using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-467 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

OPTIMIZE DEMONSTRATION

SSS 1-3 | Sensor & Monitoring Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for calibration and monitoring quality.

DEMONSTRATE

Demonstrate how to optimize using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

STRESS-TEST DEMONSTRATION

SSS 1-3 | Sensor & Monitoring Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for calibration and monitoring quality.

DEMONSTRATE

Demonstrate how to stress-test using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

RECOMMEND DEMONSTRATION

SSS 1-3 | Sensor & Monitoring Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for calibration and monitoring quality.

DEMONSTRATE

Demonstrate how to recommend using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-470 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

FRAME DEMONSTRATION

SSS 1-3 | Smart Irrigation Demonstrator

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for feedback control.

DEMONSTRATE

Demonstrate how to frame using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-471 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

DESIGN DEMONSTRATION

SSS 1-3 | Smart Irrigation Demonstrator

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for feedback control.

DEMONSTRATE

Demonstrate how to design using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-472 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

INSTRUMENT DEMONSTRATION

SSS 1-3 | Smart Irrigation Demonstrator

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for feedback control.

DEMONSTRATE

Demonstrate how to instrument using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-473 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

CALIBRATE DEMONSTRATION

SSS 1-3 | Smart Irrigation Demonstrator

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for feedback control.

DEMONSTRATE

Demonstrate how to calibrate using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

MODEL DEMONSTRATION

SSS 1-3 | Smart Irrigation Demonstrator

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for feedback control.

DEMONSTRATE

Demonstrate how to model using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-475 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

VALIDATE DEMONSTRATION

SSS 1-3 | Smart Irrigation Demonstrator

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for feedback control.

DEMONSTRATE

Demonstrate how to validate using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-476 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

EVALUATE DEMONSTRATION

SSS 1-3 | Smart Irrigation Demonstrator

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for feedback control.

DEMONSTRATE

Demonstrate how to evaluate using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-477 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

OPTIMIZE DEMONSTRATION

SSS 1-3 | Smart Irrigation Demonstrator

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for feedback control.

DEMONSTRATE

Demonstrate how to optimize using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-478 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

STRESS-TEST DEMONSTRATION

SSS 1-3 | Smart Irrigation Demonstrator

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for feedback control.

DEMONSTRATE

Demonstrate how to stress-test using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

RECOMMEND DEMONSTRATION

SSS 1-3 | Smart Irrigation Demonstrator

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for feedback control.

DEMONSTRATE

Demonstrate how to recommend using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-480 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

FRAME DEMONSTRATION

SSS 1-3 | Research & Design Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for engineering design decisions.

DEMONSTRATE

Demonstrate how to frame using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-481 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

DESIGN DEMONSTRATION

SSS 1-3 | Research & Design Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for engineering design decisions.

DEMONSTRATE

Demonstrate how to design using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

INSTRUMENT DEMONSTRATION

SSS 1-3 | Research & Design Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for engineering design decisions.

DEMONSTRATE

Demonstrate how to instrument using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-483 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

CALIBRATE DEMONSTRATION

SSS 1-3 | Research & Design Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for engineering design decisions.

DEMONSTRATE

Demonstrate how to calibrate using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-484 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

MODEL DEMONSTRATION

SSS 1-3 | Research & Design Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for engineering design decisions.

DEMONSTRATE

Demonstrate how to model using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-485 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

VALIDATE DEMONSTRATION

SSS 1-3 | Research & Design Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for engineering design decisions.

DEMONSTRATE

Demonstrate how to validate using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-486 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

EVALUATE DEMONSTRATION

SSS 1-3 | Research & Design Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for engineering design decisions.

DEMONSTRATE

Demonstrate how to evaluate using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-487 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

OPTIMIZE DEMONSTRATION

SSS 1-3 | Research & Design Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for engineering design decisions.

DEMONSTRATE

Demonstrate how to optimize using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

STRESS-TEST DEMONSTRATION

SSS 1-3 | Research & Design Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for engineering design decisions.

DEMONSTRATE

Demonstrate how to stress-test using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

RECOMMEND DEMONSTRATION

SSS 1-3 | Research & Design Garden

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for engineering design decisions.

DEMONSTRATE

Demonstrate how to recommend using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-490 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

FRAME DEMONSTRATION

SSS 1-3 | Integrated Food Systems Laboratory

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for integrated system behavior.

DEMONSTRATE

Demonstrate how to frame using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-491 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

DESIGN DEMONSTRATION

SSS 1-3 | Integrated Food Systems Laboratory

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for integrated system behavior.

DEMONSTRATE

Demonstrate how to design using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-492 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

INSTRUMENT DEMONSTRATION

SSS 1-3 | Integrated Food Systems Laboratory

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for integrated system behavior.

DEMONSTRATE

Demonstrate how to instrument using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

CALIBRATE DEMONSTRATION

SSS 1-3 | Integrated Food Systems Laboratory

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for integrated system behavior.

DEMONSTRATE

Demonstrate how to calibrate using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

MODEL DEMONSTRATION

SSS 1-3 | Integrated Food Systems Laboratory

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for integrated system behavior.

DEMONSTRATE

Demonstrate how to model using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-495 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

VALIDATE DEMONSTRATION

SSS 1-3 | Integrated Food Systems Laboratory

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for integrated system behavior.

DEMONSTRATE

Demonstrate how to validate using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

EVALUATE DEMONSTRATION

SSS 1-3 | Integrated Food Systems Laboratory

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for integrated system behavior.

DEMONSTRATE

Demonstrate how to evaluate using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

OPTIMIZE DEMONSTRATION

SSS 1-3 | Integrated Food Systems Laboratory

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for integrated system behavior.

DEMONSTRATE

Demonstrate how to optimize using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

STRESS-TEST DEMONSTRATION

SSS 1-3 | Integrated Food Systems Laboratory

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for integrated system behavior.

DEMONSTRATE

Demonstrate how to stress-test using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.

RECOMMEND DEMONSTRATION

SSS 1-3 | Integrated Food Systems Laboratory

SET UP

Specify objective, system boundary, performance metric, assumptions, constraints, acceptance criteria and QC for integrated system behavior.

DEMONSTRATE

Demonstrate how to recommend using the protocol, instrument/reference IDs, traceable data and validation/QC requirements.

ASK

What decision must this evidence support? Which assumption matters? How is quality checked? What trade-off limits optimization?

TD-500 | OBSERVE -> CHECK -> CLOSE

OBSERVE

Check reproducibility, calibration/reference evidence, calculations/models, uncertainty, validation and technical reasoning.

CHECK LEARNING

Student produces technically defensible evidence that supports a design, operational or optimization decision.

CLOSE

Document limitations and corrective actions, secure systems, and define the next validation, sensitivity or prototype test.

LINK THE LEARNING

Record evidence in the GrowMeal journal -> assign the related Garden Mission Card -> use the matching practical/data/problem-solving assessment item.